Executive Summary

Aviation education can greatly benefit public schools by improving STEM learning, helping students get ready for future jobs, and opening doors to careers in high-demand fields. However, most Wisconsin public schools do not offer aviation programs. This is not due to a lack of interest, but rather because of ongoing financial and structural challenges that limit schools' ability to implement and sustain aviation instruction.

A statewide survey found that the majority of schools do not have any formal aviation programs, even those with STEM options. The main reason, cited by over three-quarters of respondents, is not having enough funding. Costs for equipment, teaching materials, teacher training, and keeping programs running are seen as major hurdles. These problems affect schools in cities, suburbs, and rural areas alike, showing that the issue is widespread. Where aviation education does exist, it is usually brief or informal, with only a small fraction of schools offering regular aviation instruction.

Despite these challenges, many educators strongly support the idea of aviation education. They see its value in making students more interested in learning, exposing them to different careers, and preparing them for today's workforce. This brief offers practical, low-cost ideas, such as adding aviation topics to current lessons, using short modules, creating partnerships with aviation groups, and sharing resources, that can help schools provide aviation education without high costs.

Future of Flight Equity aims to close the gap between interest and access by sharing data and strategies that make aviation education possible and fair for all students in Wisconsin at a time when workforce readiness and STEM equity are increasingly important.

The Access Gap in Aviation Education

The presence of aviation education within Wisconsin's schools varies widely, with some institutions offering limited exposure while many have no programs at all. This inconsistency is primarily the result of financial challenges, rather than differences in geography or a lack of student interest. Results from a statewide survey confirm that the single greatest barrier to implementing aviation-related courses, activities, or extracurriculars is insufficient funding.

Among the schools that participated in the survey, most reported that they do not have any official aviation programs. This is true even in schools where there are established pathways for science, technology, engineering, and mathematics (STEM). In many of these cases, aviation topics simply are not included in the STEM curriculum. Educators overwhelmingly stated that this exclusion is not because aviation is seen as unimportant or irrelevant; instead, it is due to practical limitations, especially the lack of adequate funding and resources.

When schools were asked in more detail about why they do not offer aviation instruction, a clear majority pointed directly to financial issues as the main cause. The costs involved cover a range of needs: purchasing specialized equipment, acquiring instructional materials, training teachers, and maintaining program sustainability. All kinds of schools, including urban, suburban, and rural, face these financial obstacles. Similarly, schools serving different types of student populations experience the same challenges, suggesting that the problem is widespread across the state.

In cases where students do have some exposure to aviation, it is usually through short-term or informal means. For example, a school might invite a guest speaker, organize a single seminar, or offer an extracurricular club. Only a small percentage of schools reported that they provide ongoing, structured aviation instruction as a regular part of their curriculum. This highlights how financial constraints make it difficult for schools to both start and sustain robust aviation programs.

Feedback from educators indicates that aviation education is widely seen as valuable. Respondents believe that offering aviation instruction could increase student engagement in STEM subjects, help students discover new career paths, and better prepare them for future employment. Despite this recognition of its importance, most schools struggle to find affordable ways to introduce aviation programming. Many respondents said they lack access to cost-effective models or external funding sources, making finances the primary hurdle.

Overall, the evidence suggests that the reason for limited aviation education in Wisconsin public schools is not a lack of regional equity or institutional interest. Instead, the problem is rooted in a structural funding gap that affects nearly every type of school at a time when the aviation and aerospace workforce continues to grow. Addressing this issue will require solutions that help schools get started with aviation programs, lower the initial and ongoing expenses, and offer practical, affordable models for sustainable instruction. By making these changes, it will become more feasible for Wisconsin schools to provide aviation education to a broader range of students.

Why Aviation Belongs in Public Education

Aviation education offers a strong way to connect what students learn in school with real-life skills, job preparation, and future career opportunities. This field draws on knowledge from physics, math, engineering, technology, and problem-solving, making it a great way to teach STEM subjects in K–12 classrooms.

Often, students only see STEM topics in theory. Aviation gives them a real-world setting to apply these ideas. Subjects like motion, energy, engineering, navigation, and data analysis are key parts of aviation and match well with current science and math standards. By learning these topics through aviation, students can understand them better and stay more engaged.

Aviation education also helps students learn about different career paths early on. The aviation and aerospace industries offer many good jobs, such as engineering, aircraft maintenance, air traffic control, and roles in new fields like advanced air mobility and space. Introducing students to these options helps them make informed choices about their futures, especially those who might not hear about aviation at home.

Making aviation part of public education also helps ensure that all students, regardless of their background, have a fair chance to learn about these careers. Schools are the best place to reach students who may not have other ways to discover aviation. Adding aviation topics into existing courses or after-school clubs is possible without changing the main school curriculum, making it easy to fit aviation education into current school structures.

Overall, aviation education supports key goals of public education: strong academics, job readiness, and equal access to opportunities. When used well, it becomes a valuable part of STEM education that helps prepare students for the modern workforce.

Barriers Schools Face

Although aviation education is valued, public schools encounter several real challenges when trying to offer these programs. Most of these difficulties come from the way schools are structured, not from a lack of interest or support.

The biggest challenge is limited funding. Many school districts have tight budgets and must focus on essential classes, staff, and required programs. This leaves little room for adding new subjects like aviation. Aviation programs are often seen as too expensive because of costs for equipment, materials, teacher training, and ongoing support. Even when less expensive options are available, schools may not have the extra resources to try them out.

Another issue is finding staff with the right experience. Few teachers have a background in aviation or aerospace, and there is often not enough time for additional training. Teachers already handle many responsibilities, so adding a new subject can be difficult without extra support.

There are also administrative and logistical barriers. Schools need to consider scheduling, approval processes, and how new content fits with state standards. Even offering aviation as an elective requires planning and coordination. Without clear guidelines or help from outside sources, schools may see aviation programs as too complicated to manage.

Finally, many schools are not aware of possible partnerships or funding sources. They may not know where to look for help from nonprofits, aviation organizations, or industry partners. This lack of information makes it even harder to find the support they need.

To make aviation education more available, it is important to address these challenges by reducing costs, easing the workload for teachers, and providing simple, clear steps for starting programs. Tackling these issues directly will help more students in Wisconsin gain access to aviation education.

Actionable, Scalable Solutions

To make aviation education more available in Wisconsin public schools, schools need practical solutions that fit their current budgets and structures. Instead of relying on expensive or specialized programs, it is helpful to focus on lowering barriers, building partnerships, and including aviation topics within classes that already exist. The strategies below offer realistic ways for schools to add aviation education while keeping costs and risks low.

Schools can immediately implement aviation topics in current STEM classes. Schools do not need to create separate aviation courses. Instead, they can use aviation examples and projects in physics, math, engineering, or technology lessons. Topics such as motion, energy, navigation, and system design are naturally related to aviation and fit well into existing courses. This method makes lessons more engaging without needing extra staff or major curriculum changes.

A low-risk pilot model includes short, low-cost modules. Schools can use brief units, lasting a week or two, to introduce students to aviation and related careers by using digital resources, simulations, or hands-on problem-solving activities. These modules do not require expensive equipment and can be added to regular class periods or enrichment times.

Working with local partners can also help schools offer aviation education. Partnerships with aviation groups, flight schools, museums, universities, or industry professionals can bring in guest speakers, mentors, or opportunities for students to visit facilities. These collaborations can provide real-world experiences for students at little or no cost and can be adapted for any community.

Extracurricular activities offer another way to start. After-school clubs, seminars, or short workshops need fewer approvals and allow schools to test aviation programs before making bigger commitments. These options can help schools see if there is enough interest before expanding further.

Finally, sharing resources between schools can make starting aviation programs easier. Having a central collection of lesson plans, guides, partner contacts, and funding sources would help schools get started without repeating work. This shared resource could be updated by teachers or outside organizations as new materials become available.

Together, these approaches focus on affordability, flexibility, and long-term success. By working with what schools already have and building partnerships, aviation education can become a realistic and valuable part of public education in Wisconsin.

Call to Action

The information in this brief shows that aviation education is limited in Wisconsin public schools mainly due to funding and access issues, not because of a lack of interest or importance. To overcome these challenges, schools, policymakers, aviation groups, and community partners need to work together to make aviation education possible and lasting for students across the state.

Schools are encouraged to try affordable and flexible ways to offer aviation education. This could include adding aviation topics to current STEM classes, using short-term units, or starting extracurricular clubs and activities. By testing options that fit with what they already have, schools can give more students access to aviation without taking on large costs.

Aviation groups, nonprofits, and industry partners are important in lowering costs and supporting schools. They can help by offering guest lessons, mentoring, curriculum help, or access to aviation facilities. Strong partnerships between schools and the aviation community are key to reaching more students.

Policymakers and school leaders can help by raising awareness, encouraging the sharing of resources, and finding ways to fund aviation programs fairly. Even modest, targeted investments can lead to meaningful results if they are designed to grow over time.

Future of Flight Equity aims to bring together data, schools, and partners to help expand aviation education in Wisconsin. By working together and sharing resources, everyone involved can make sure students have access to aviation education based on their interests and abilities, not on their financial situation. The opportunity now is to move aviation education from a limited offering to a sustainable pathway for the next generation.